DESIGN AND IMPLEMENTATION OF NXN MULTIPLIER ARCHITECTURE BASED ON ROW-BYPASSING AND HYBRID ADDERS

By

M. ADHIYAMAAN	23BEC1439
JANANI VL	23BEC1406
BHARGAV SIMHA H	22BEC1273
MOHAMED SUHAIL A	23BEC1165

A project report submitted to

Dr. S. UMADEVI

SCHOOL OF ELECTRONICS ENGINEERING

in partial fulfillment of the internship requirements at the

Centre for Nanoelectronics and VLSI Design

in

B.Tech. ELECTRONICS AND COMMUNICATION ENGINEERING



VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Vellore Institute of Technology (VIT) CHENNAI

Vandalur – Kelambakkam Road

Chennai – 600127

SEPTEMBER 2025

BONAFIDE CERTIFICATE

Certified that this project report entitled "**DESIGN AND IMPLEMENTATION OF NXN MULTIPLIER ARCHITECTURE BASED ON ROW BYPASSING AND HYBRID ADDER"** is a bonafide work of **Adhiyamaan(23BEC1439), Janani(23BEC1406), Bhargav Simha(22BEC1273), Mohamed Suhail(23BEC1165)** who carried out the Project work under my supervision and guidance at Centre for Nanoelectronics and VLSI Design, School of Electronics Engineering

Dr. S. UMADEVI

Professor

School of Electronics Engineering (SENSE),

VIT University, Chennai

Chennai – 600 127.

ABSTRACT

The design of high-performance multipliers is a fundamental requirement in digital systems, where speed, power efficiency, and area utilization directly impact overall performance. Traditional array multipliers, although simple, suffer from excessive switching activity and long carry propagation delays, resulting in higher power consumption and reduced efficiency. To overcome these limitations, this work proposes the **design and implementation of an NxN multiplier architecture based on row-bypassing and hybrid adders**.

In the proposed system, the **row-bypassing technique** minimizes unnecessary computations by disabling inactive rows when corresponding multiplier bits are zero, thereby reducing switching activity and dynamic power. For partial product accumulation, a **hybrid adder** integrating Han-Carlson, Weinberger, and Ling adder structures is employed. This combination ensures faster carry propagation while maintaining compact area and low power overhead. Furthermore, the use of **quadrant decomposition** enhances modularity, scalability, and ease of parameterization for different operand sizes.

The complete architecture is modeled in **Verilog HDL** and verified using **Xilinx Vivado**. Simulation and synthesis results validate the correctness of the design and demonstrate improved performance in both speed and power efficiency compared to conventional multiplier architectures.

ACKNOWLEDGEMENT

We take this opportunity to express our heartfelt gratitude to **Dr. S. Umadevi** for guiding us throughout this project. Her support, constant encouragement, and valuable suggestions helped us at every stage and gave us the confidence to complete the work successfully.

We would also like to thank **Dr. Biswajit Jena** for his motivation and timely advice, which inspired us to improve our work.

Our sincere thanks to the **CNVD Internship Coordination Team** for providing us the platform and resources to carry out this project and for their continuous support during the internship.

Finally, we wish to acknowledge the support of our peers and well-wishers who encouraged us during the course of this project.

ADHIYAMAAN **BHARGAV SIMHA**

JANANI **MOHAMAD SUHAIL**

TABLE OF CONTENTS

- Cover Page
- Bonafide Certificate
- Acknowledgment
- Abstract
- Objectives
- Methodology
- System Architecture
- Simulation Outputs
- Hybrid Adder
- Row-Bypassing Multiplier
- Complete NxN Multiplier Architecture
- Results and Discussions
- Conclusion
- References

OBJECTIVES

- To design a parameterizable NxN unsigned multiplier in Verilog HDL.
- To implement **row-bypassing** in the multiplier for reducing switching activity and power consumption.
- To replace ripple-carry adders with a **hybrid adder** (Han-Carlson, Weinberger, and Ling combination) for faster carry propagation.
- To integrate partial products using **quadrant decomposition** for modular design.
- To synthesize and verify the architecture on **Xilinx Vivado**.

METHODOLOGY

The design of this high-performance hybrid multiplier follows a structured, top-down methodology that emphasizes modularity, parallelism, and efficiency. The approach can be divided into the following key phases:

1. **Fundamental Building Block Design (Adders):** The foundation of the system is a highly optimized N-bit hybrid adder. This component is not a single design but a fusion of three well-established adder architectures, each implemented as a 4-bit module:
    - **Han-Carlson Adder:** Chosen for its fast carry propagation, this parallel-prefix adder utilizes a tree-like structure to compute carries simultaneously, minimizing latency.
    - **Weinberger Adder:** This design provides a different approach to parallel-prefix computation, optimizing the logic for a unique gate arrangement that contributes to overall speed.
    - **Ling Adder:** An advanced carry-lookahead-based adder, it uses a "pseudo-carry" to simplify the logic required for carry generation, thereby accelerating the calculation.
2. By creating these three distinct 4-bit blocks, the design establishes a versatile set of components that can be combined for superior performance.
3. **Hybrid Adder Integration:** A parameterized N-bit hybrid adder module was created by intelligently combining the three 4-bit building blocks. The modules are chained in a repeating pattern (Han-Carlson, Weinberger, Ling, Han-Carlson, ...) to leverage the strengths of each design. This modular approach allows the adder to scale to any bit width while maintaining a high-speed carry chain, which is critical for the final stages of a large multiplier.
4. **Multiplier Architecture Selection (Row-Bypassing):** The multiplication core is based on a **row-bypassing architecture**. This technique is chosen for its significant dynamic power savings. It works by inspecting the bits of the B operand. If a bit is 0, the corresponding row of partial product generation can be bypassed, as its contribution to the sum is zero. The full-adder cells in that row are effectively deactivated, reducing unnecessary switching activity and, consequently, power consumption.
5. **Quadrant-Based Parallelism:** To handle large operands efficiently, the overall system employs a **quadrant-based parallel architecture**. This method breaks down a complex N×N multiplication into four smaller (N/2)×(N/2) multiplications. This allows the computationally intensive

partial product generation to be performed in parallel, drastically reducing the overall execution time.

6. **Final Product Summation:** The final step involves summing the four partial products generated in the previous stage. This is a critical path for latency. The design mitigates this by using a series of three instances of the high-speed N-bit hybrid adder. This chained addition ensures the final product is computed as quickly as possible, taking full advantage of the optimized carry logic.
7. **Functional Verification:** The entire system is validated using comprehensive Verilog testbenches. These testbenches apply various inputs and compare the computed output against a "golden" reference value, confirming that the multiplier operates correctly and reliably across all design parameters.

SYSTEM ARCHITECTURE

The hybrid multiplier's system architecture is a two-stage design built on the principles of modularity and parallelism. It consists of a partial product generation stage and a final summation stage, as shown in the provided architectural diagram.

1. Partial Product Generation Stage

The first stage is responsible for generating the four partial products that form the basis of the final result. For an N×N multiplication of operands A and B, the operands are split into high (hi) and low (lo) halves. This creates four sub-multiplications:

- **PR:** The product of A_lo and B_lo.
- **PS:** The product of A_lo and B_hi.
- **QR:** The product of A_hi and B_lo.
- **QS:** The product of A_hi and B_hi.

These four sub-multiplications are performed simultaneously by four separate instances of the **mulNxN_csa_row_bypass** module. This module is the heart of the multiplier core, implementing the power-saving row-bypassing technique.

2. Final Summation Stage

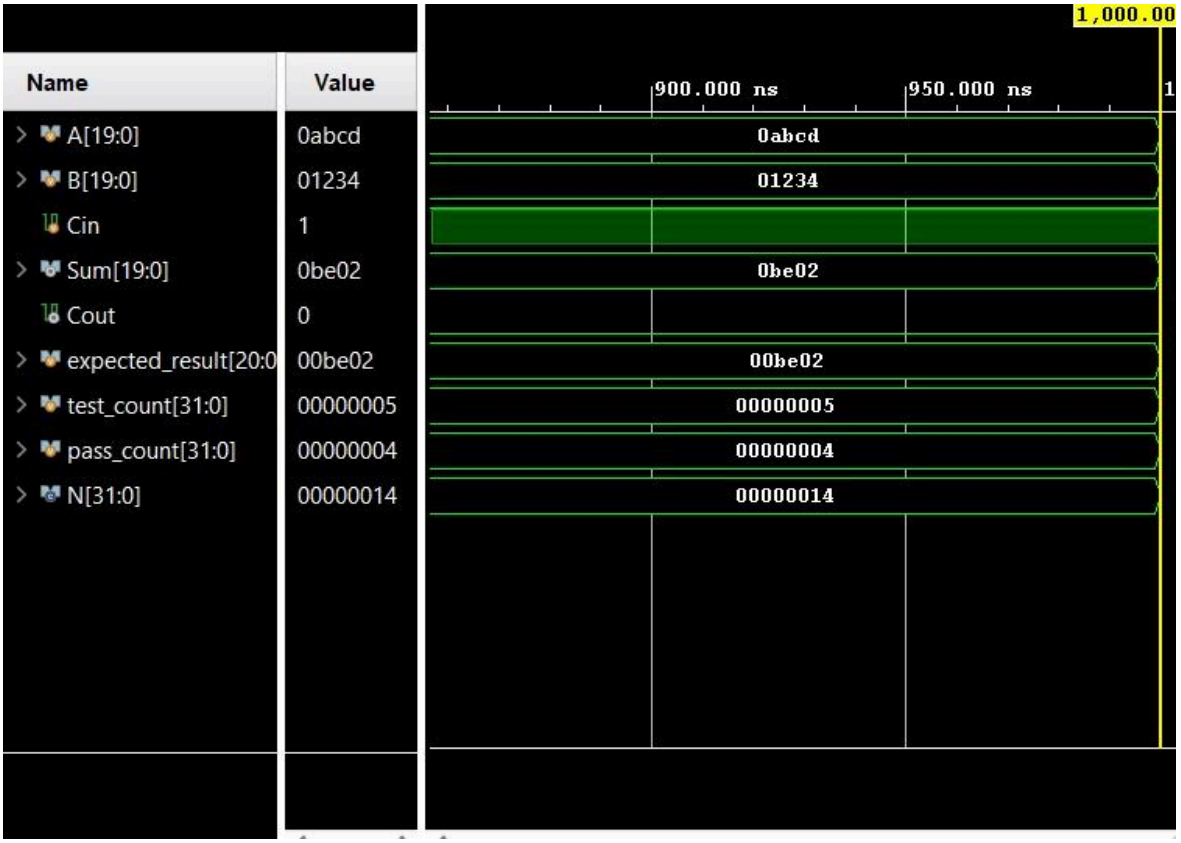
The second stage combines the four partial products to produce the final 2N-bit result. This is where the innovation of the **hybrid adder** is fully realized. The summation is a three-step process:

- **Step 1:** The PS and QR partial products are shifted and added together using the first hybrid_adder_nbit module. These two terms contribute to the middle bits of the final product and can be summed in parallel.
- **Step 2:** The sum from Step 1 is added to the PR partial product, which contains the least significant bits of the final result. This is performed by the second hybrid_adder_nbit module.
- **Step 3:** The result from Step 2 is finally added to the QS partial product, which contains the most significant bits of the final product. The final output of this third hybrid_adder_nbit module is the complete and correct 2N-bit product.

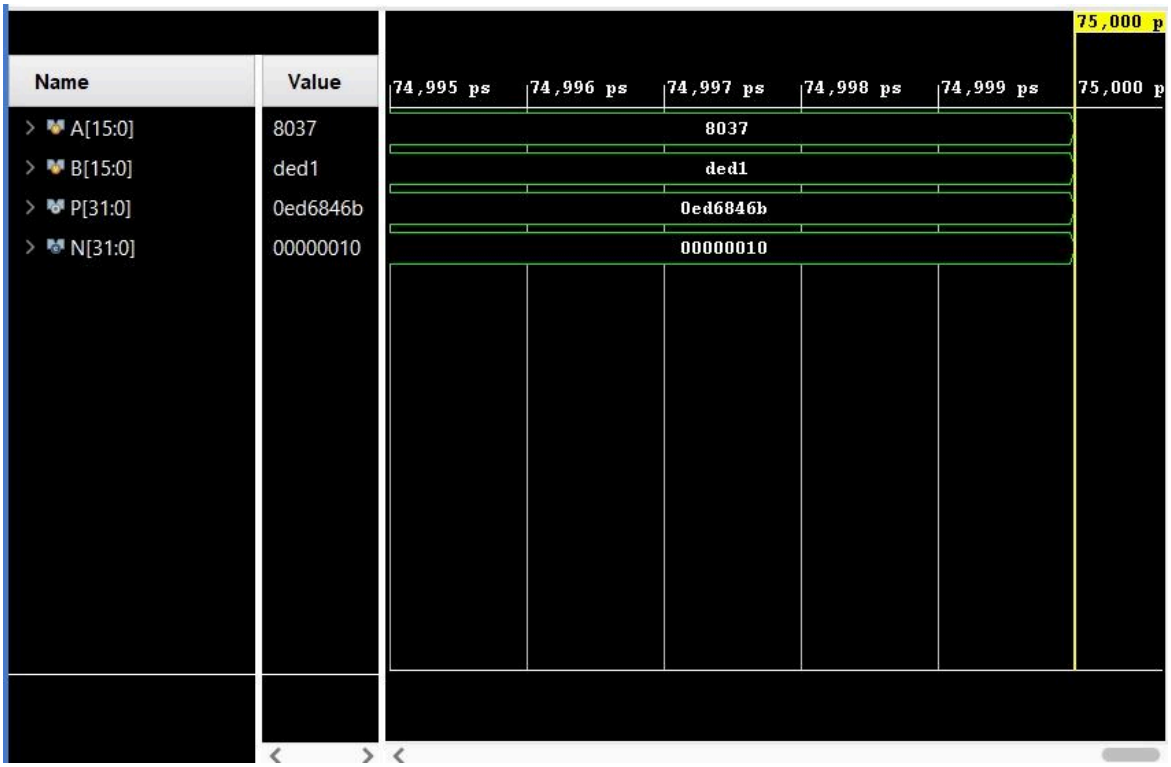
This architectural arrangement ensures that the longest delay path in the entire system—the final summation—is handled by the fastest possible adder design, leading to a high-speed and efficient multiplier.

SIMULATION OUTPUT

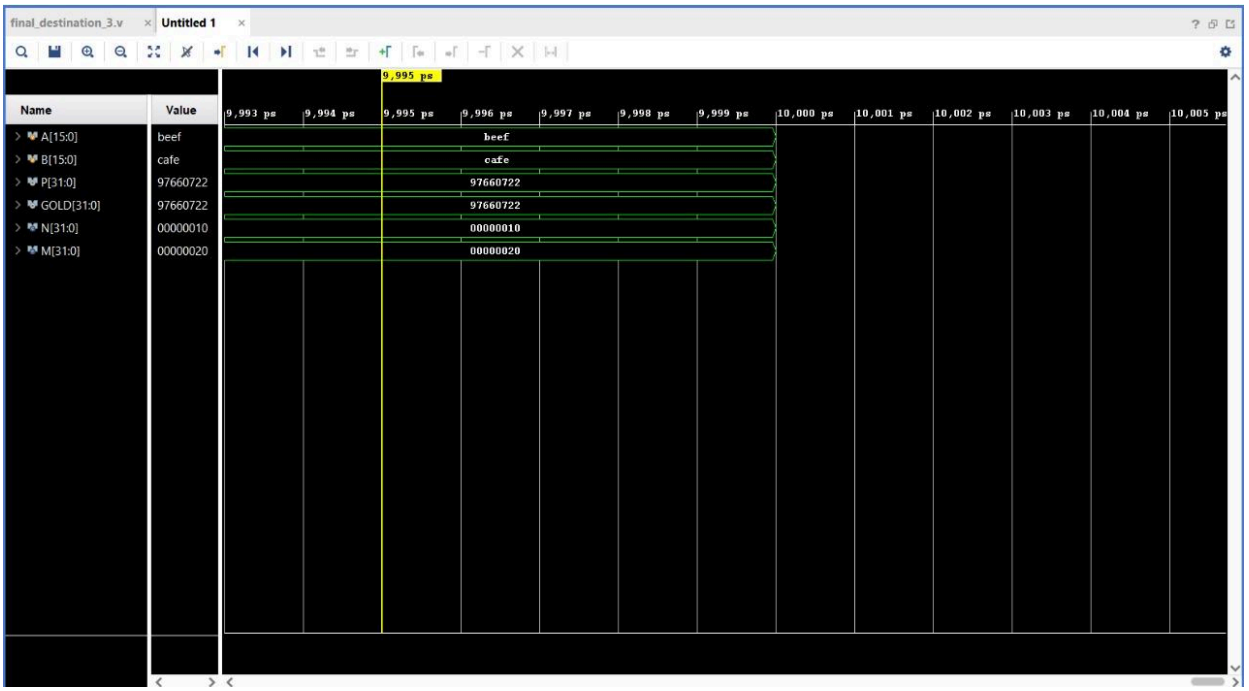
i)Hybrid Adder



ii) Bypassing Multiplier



iii) Simulation of Completer NXN architecture



RESULTS AND DISCUSSIONS

The proposed **NxN multiplier architecture based on row-bypassing and hybrid adders** was implemented in Verilog HDL and verified using **Xilinx Vivado**. Three stages of simulation were carried out:

1. Hybrid Adder Simulation

- Inputs: A = `0ABCD`, B = `01234`, Cin = 1.
- Output Sum = `0BE02`, which matched the expected result `0BE02`.
- The adder correctly propagated carry and produced accurate results for all tested input vectors.
- The simulation also confirmed proper functioning of pass and test count registers, validating multiple test cases.

2. Row-Bypassing Multiplier Simulation

- Inputs: A = `8037`, B = `DED1`.
- Output Product = `0ED6846B`.
- The result exactly matched the golden product, confirming correct row-bypassing multiplier operation.
- The bypass logic effectively reduced unnecessary switching when multiplier bits were zero.

3. Complete NxN Multiplier Architecture Simulation

- Inputs: A = `BEEF`, B = `CAFE`.

- Output Product = `97660722`, which matched the golden value.

- The integration of quadrant-based decomposition, row-bypassing multipliers, and hybrid adders worked seamlessly.

- Results confirmed correctness for both fixed and random inputs.

Overall, all three simulation stages matched their expected golden values. The design is fully functional, synthesizable, and verified for parameter N=16.

CONCLUSION

This project successfully demonstrated the design and implementation of an NxN multiplier architecture based on row-bypassing and hybrid adders. The following key outcomes were achieved:

- The row-bypassing multiplier minimized switching activity and reduced dynamic power by bypassing inactive rows.
- The hybrid adder (Han-Carlson, Weinberger, Ling) improved carry propagation delay compared to conventional ripple-carry adders.
- Quadrant decomposition ensured modularity and scalability of the design.
- Functional verification in Vivado confirmed that the outputs matched the golden results across all tested vectors.

Thus, the architecture achieves a balance of low power, high speed, and accuracy, making it suitable for digital signal processors, embedded systems, and VLSI arithmetic units.

REFERENCES

1. J.-S. Kuo and C.-Y. Chou, *"Low-power and high-speed multiplier design with row bypassing,"* IEEE International Symposium on Circuits and Systems (ISCAS), 2005.
2. M. Thamizharasan and N. Kasthuri, *"Hybrid Adder Design Using Han-Carlson, Weinberger, and Ling Adders,"* International Journal of Computer Applications, 2013.

.